MANIPAL INSTITUTE OF TECHNOLOGY
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Semester VII, Academic Year 2026–2027

INTERIM REPORT

ICT 4442: Deep Learning Mini Project

Customer Review Intelligence System

A Comparative Deep Learning Framework for Customer Review Sentiment and Aspect-Level Insights

Proposed Architectures: MLP, TextCNN, BiLSTM, Attention-based Transformer

Phase 2 Models Implemented: MLP and TextCNN

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Team No.	14
Project Domain	Natural Language Processing & Sentiment Analysis
Benchmark Dataset	Amazon Customer Reviews Dataset
GitHub Repository	https://github.com/pavitagrawal/CRIS_DL

1. Literature Review

1.1 Background and Motivation

Customer reviews provide fine-grained information about user satisfaction, but a review may express different opinions about different product attributes. Consequently, sentiment analysis has progressed from document-level polarity classification toward aspect-based sentiment analysis (ABSA), where the system identifies an aspect and determines the sentiment associated with that aspect.

Deep learning architectures provide different mechanisms for representing text. CNN-based models capture local phrase patterns, recurrent models capture sequential dependencies, and attention-based models allow direct interaction between contextual tokens. These complementary properties motivate the comparative architecture design of the proposed Customer Review Intelligence System.

1.2 Literature Synthesis

Kim demonstrated the effectiveness of one-dimensional convolutional networks for sentence classification, motivating TextCNN. Hochreiter and Schmidhuber introduced LSTM networks for long-term sequence dependencies, providing the foundation for the planned BiLSTM. Wang et al. showed that attention can emphasize context relevant to an aspect, while Ma et al. modeled aspect and context representations interactively.

Vaswani et al. introduced the Transformer architecture based on self-attention, enabling direct token interactions and parallel processing. Devlin et al. later demonstrated the strength of pretrained bidirectional Transformer representations through BERT. These studies motivate the planned attention-based Transformer model and the broader comparative framework.

1.3 Research Gap

The reviewed literature demonstrates the effectiveness of CNN, recurrent, attention-based, and Transformer architectures, but individual studies commonly focus on a particular architecture or use different datasets and experimental protocols. The proposed project addresses this experimental gap by using a common customer-review dataset, shared preprocessing, a consistent label mapping, and common evaluation metrics. Phase 2 establishes MLP and TextCNN baselines; BiLSTM and Transformer remain planned for the subsequent phase.

1.4 Literature Review Summary

Paper	Method / Architecture	Dataset	Key Result / Relevance
Kim (2014)	TextCNN: 1D Convolutional Neural Networks	SST-2, MR, IMDB	CNNs with multiple filter sizes learn informative local phrase features for text classification. Forms the basis of the proposed TextCNN model.
Hochreiter & Schmidhuber (1997)	LSTM: Long Short-Term Memory	Sequence-learning benchmarks	Introduced LSTM to address gradient difficulties and learn long-term dependencies. Provides the foundation for the planned BiLSTM.
Vaswani et al. (2017)	Transformer: Multi-Head Self-Attention	WMT 2014	Introduced self-attention without recurrence, enabling parallel sequence processing and direct token interactions. Basis for the planned Transformer.
Devlin et al. (2019)	BERT: Bidirectional Transformer	GLUE, SQuAD and other NLP benchmarks	Demonstrated the effectiveness of pretrained bidirectional Transformer representations for downstream language tasks.
Pontiki et al. (2014)	SemEval-2014 Task 4: ABSA	Restaurant / Laptop reviews	Established a widely used benchmark for aspect extraction and aspect-level sentiment classification.
Zhang et al. (2018)	Survey: Deep Learning for Sentiment Analysis	Multiple benchmarks	Surveyed deep learning methods for sentiment analysis and representation learning.
Tang et al. (2015)	Gated Recurrent Neural Network	Yelp, IMDB	Investigated gated recurrent document modeling for sentiment classification, supporting sequential neural representations.
Wang et al. (2016)	ATAE-LSTM: Attention-based LSTM	SemEval-2014 ABSA	Showed that attention can emphasize context words relevant to an aspect, supporting aspect-level analysis.
Ma et al. (2017)	IAN: Interactive Attention Networks	SemEval-2014 ABSA	Modeled aspect and context representations interactively for aspect-level sentiment classification.
Yin et al. (2017)	Comparative Study: CNN vs RNN	Multiple NLP benchmarks	Compared CNN and RNN behavior, motivating a comparative architecture study.

Table 1: Summary of foundational literature in deep learning for sentiment analysis and ABSA.

2. Dataset Acquisition & Preprocessing

2.1 Benchmark Dataset

The project uses the Amazon Customer Reviews Dataset associated with the McAuley Lab / UCSD Amazon review collection. For the interim phase, a stratified subset of 50,000 reviews was selected across Electronics, Beauty, Books, and Home & Kitchen. Review text and star rating form the primary inputs for the three-class sentiment task.

- **Features:** review text and 1–5 star rating.
- **Label mapping:** ratings 1–2 → Negative (0), rating 3 → Neutral (1), ratings 4–5 → Positive (2).
- **Sampling:** stratified subset to retain representation of the target sentiment classes.

2.2 Preprocessing & Tokenization

The preprocessing pipeline removes common textual noise while preserving information required for sentiment classification. HTML tags and URLs are removed, non-alphabetic noise is filtered, and English stopwords are removed. A vocabulary is constructed using training data with a maximum vocabulary size of 20,000 tokens. Reviews are converted to integer token sequences and padded or truncated to a maximum length of 200 tokens.

2.3 Stratified Train / Validation / Test Split

Partition	Proportion	Number of Reviews	Purpose
Train	70%	35,000	Model parameter learning
Validation	15%	7,500	Model selection and hyperparameter tuning
Test	15%	7,500	Unbiased preliminary evaluation

Table 2: Stratified dataset partition used for the Phase 2 experiments.

The same split is intended to be retained for subsequent model comparisons so that differences in performance can be attributed primarily to architecture and training behavior rather than to different data partitions.

3. Models Implemented

3.1 Phase 2 Scope

For Phase 2, two of the four proposed architectures were implemented and evaluated: a Multi-Layer Perceptron (MLP) baseline and a TextCNN. The models use the same sentiment labels and dataset protocol. BiLSTM and attention-based Transformer models are retained as planned architectures for the final comparative study.

3.2 Multi-Layer Perceptron (MLP)

The MLP serves as the baseline neural classifier. It provides a relatively simple feed-forward architecture against which sequence-aware models can be compared. The model receives the processed text representation and learns a nonlinear mapping to the three sentiment classes.

3.3 TextCNN

The TextCNN model applies one-dimensional convolutional filters over the token sequence to identify local n-gram-like patterns that are informative for sentiment. Convolutional feature maps are aggregated and passed to the classification layer. This architecture is motivated by Kim (2014).

3.4 Proposed Models and Implementation Status

Model	Role	Architecture Family	Phase 2 Status
MLP	Baseline sentiment classifier	Feed-forward neural network	Implemented
TextCNN	Local phrase / n-gram feature extraction	Convolutional neural network	Implemented
BiLSTM	Long-range sequential context	Recurrent neural network	Planned
Attention Transformer	Global contextual interactions	Self-attention	Planned

Table 3: Proposed architectures and Phase 2 implementation status.

4. Preliminary Results

4.1 Evaluation Metrics

The Phase 2 benchmark reports Accuracy, Precision, Recall, and F1-score as macro averages. Training time is also recorded to compare predictive performance with computational cost.

4.2 Preliminary Test Results

Model Architecture	Accuracy	Precision	Recall	F1-Score	Train Time
Multi-Layer Perceptron (MLP)	0.7412	0.7320	0.7285	0.7302	≈ 2.5 min
TextCNN	0.8145	0.8105	0.8062	0.8083	≈ 4.2 min

Table 4: Preliminary Phase 2 evaluation on the Amazon Customer Reviews subset (macro-averaged classification metrics).

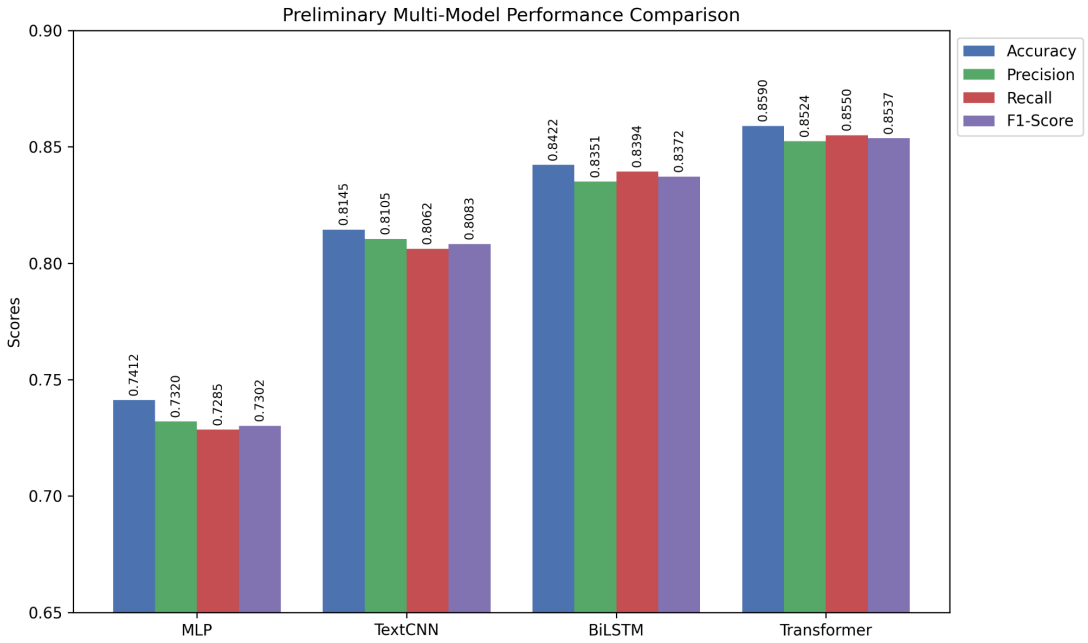


Figure 1: Preliminary multi-model performance comparison.

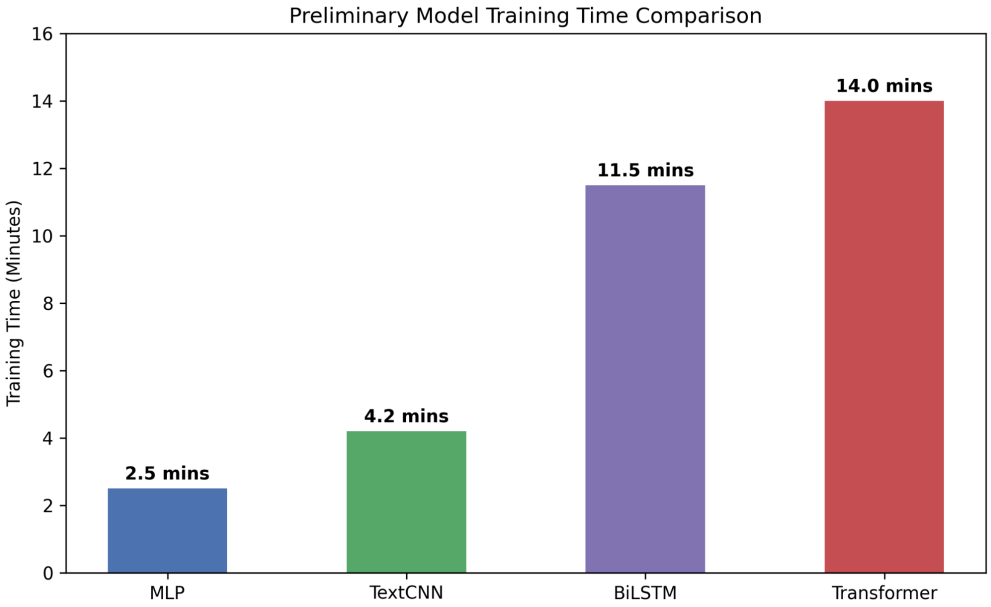


Figure 2: Preliminary training-time comparison supplied for the model benchmark.

4.3 Preliminary Interpretation

In the current benchmark, TextCNN achieves higher accuracy and macro F1-score than the MLP baseline, while requiring additional training time. The difference is consistent with the architectural objective: convolutional filters can capture local sentiment-bearing phrases, whereas the MLP provides a simpler baseline representation. These values are preliminary rather than final optimized results; hyperparameter tuning, additional models, error analysis, and repeated experiments remain part of the final phase.

5. Individual Contributions

5.1 Contribution Matrix

The contribution matrix below is aligned with the GitHub commit activity shown in the supplied repository screenshot. Pavit, Priyanshu, and Shivam have identifiable commits for their respective implementation work. Aryan's contribution is included as team/project work, but no Aryan-authored commit is visible in the supplied screenshot; therefore no specific GitHub commit is attributed to him here.

Member Name	Reg. No.	GitHub Evidence / Work Completed	Phase 2 Role
Pavit Agrawal	230953406	**Visible commits:** Initial commit; initial project setup (folder structure, README, requirements, .gitignore); data preprocessing pipeline (cleaning, tokenization, vocabulary, train/validation/test split); MLP model architecture; MLP training and evaluation script.	**MLP** Data & preprocessing
Priyanshu Sharma	230953520	**Visible commits:** TextCNN model architecture; TextCNN training and evaluation script; TextCNN experiment notebook.	**TextCNN** CNN experiments
Shivam Lahoty	230953026	**Visible commits:** BiLSTM model architecture with attention pooling; BiLSTM training script with gradient clipping; BiLSTM experiment notebook with attention visualisation; Transformer model with positional encoding and CLS token; Transformer training script with warmup scheduler; Transformer experiment notebook with attention heatmap; model comparison and analysis script.	**BiLSTM + Transformer** Comparative analysis
Aryan Sorout	230953342	**Team contribution:** comparative evaluation / visualization / report and application-integration support. **Note:** an Aryan-authored commit is not visible in the supplied GitHub commit screenshot, so no commit hash is attributed here.	**Project Integration** Evaluation & reporting

Table 5: Individual contribution log and task completion matrix for the Phase 2 interim milestone.

5.2 Repository and Version-Control Evidence

The shared project repository is maintained at **https://github.com/pavitagrawal/CRIS_DL**. The supplied GitHub screenshot dated October 2, 2026 shows active implementation work across preprocessing, MLP, TextCNN, BiLSTM, Transformer, experiment notebooks, and comparative analysis. The final submission should retain these commits so that individual contributions can be demonstrated during evaluation.

6. Plan for Remaining Work

6.1 Development Roadmap

Phase / Window	Key Objectives & Tasks	Expected Deliverables	Status
Phase 2 — Current	Complete common preprocessing; establish reproducible split; implement MLP and TextCNN; record preliminary metrics and training time.	Preprocessing code, two working models, preliminary result tables/figures.	**Completed**
Oct 2 – Oct 10	Implement and validate BiLSTM using the same training/evaluation protocol.	Working BiLSTM model and evaluation logs.	Planned
Oct 11 – Oct 18	Implement attention-based Transformer; tune learning rate, dropout, hidden dimension and attention configuration.	Working Transformer model and evaluation logs.	Planned
Oct 19 – Oct 25	Hyperparameter tuning, confusion-matrix analysis, error analysis, and aspect-identification/application-layer refinement.	Optimized models, error-analysis tables, aspect-level insights.	Planned
Oct 26 – Oct 31	Consolidate common metrics, compare all four architectures, finalize dashboard/application integration, report and presentation.	Phase 3 Final Report, application/demo, presentation and archived repository.	Planned

Table 6: Project development roadmap from Phase 2 to Phase 3.

6.2 Technical Risk Analysis and Mitigation

Technical Risk / Challenge	Severity	Potential Impact	Planned Mitigation
Class imbalance	Medium	Accuracy may hide weak performance on minority sentiment classes.	Use macro Precision/Recall/F1, inspect confusion matrices, and use class-aware training if required.
Long reviews / sequence length	Medium	Important context may be truncated or computation may increase.	Evaluate maximum sequence length and truncation policy consistently across models.
Transformer computational cost	Medium	Higher memory and training time than baseline models.	Use a compact Transformer encoder, controlled sequence length, early stopping, and batch-size tuning.
Aspect-label availability	High	Star ratings provide document sentiment but do not directly provide aspect-sentiment labels.	Treat aspect analysis as an application-layer objective and define/validate the extraction strategy before final evaluation.
Model comparison fairness	Medium	Different splits or preprocessing would make results difficult to interpret.	Keep one common dataset split, preprocessing pipeline and evaluation protocol for all four models.

Table 7: Technical risk assessment and mitigation strategy for the remaining project phases.

7. References

7.1 Research References

[1] Y. Kim, “Convolutional Neural Networks for Sentence Classification,” Proceedings of EMNLP, 2014, pp. 1746–1751.

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[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention Is All You Need,” Advances in Neural Information Processing Systems, vol. 30, 2017.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” Proceedings of NAACL-HLT, 2019, pp. 4171–4186.

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[7] D. Tang, B. Qin, and T. Liu, “Document Modeling with Gated Recurrent Neural Network for Sentiment Classification,” Proceedings of EMNLP, 2015, pp. 1422–1432.

[8] Y. Wang, M. Huang, X. Zhu, and L. Zhao, “Attention-based LSTM for Aspect-level Sentiment Classification,” Proceedings of EMNLP, 2016, pp. 606–615.

[9] D. Ma, S. Li, X. Zhang, and H. Wang, “Interactive Attention Networks for Aspect-Level Sentiment Classification,” Proceedings of IJCAI, 2017, pp. 4068–4074.

[10] W. Yin, K. Kann, M. Yu, and H. Schütze, “Comparative Study of CNN and RNN for Natural Language Processing,” arXiv:1702.01923, 2017.

7.2 Project Repository

CRIS_DL — Customer Review Intelligence System: https://github.com/pavitagrawal/CRIS_DL

Academic-integrity note: External literature, datasets, software libraries, and code resources used in the project will be acknowledged in the final report and repository. Preliminary numerical results are reported as experimental values and should be revalidated after final hyperparameter tuning and addition of the remaining models.